SETUP

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Optional settings
pd.set_option('display.max_columns', None)
```

```
# Creating a realistic sales dataset
np.random.seed(0)

n = 10000

df = pd.DataFrame({
    'Order ID': np.random.randint(10000, 20000, n),
    'Product': np.random.choice(['Laptop', 'Phone', 'Headphones', 'Monitor', 'Keyboard'], n),
    'Quantity Ordered': np.random.randint(1, 5, n),
    'Price Each': np.random.choice([500, 700, 1500, 200, 100], n),
    'Order Date': pd.date_range(start='2023-01-01', periods=n, freq='H'),
    'City': np.random.choice(['Mumbai', 'Pune', 'Delhi', 'Bangalore'], n)
})

df.head()
```

```
/tmp/ipykernel_3195/1325882224.py:11: FutureWarning: 'H' is deprecated and will be removed in a future version, please use '
  'Order Date': pd.date_range(start='2023-01-01', periods=n, freq='H'),
```

	Order ID	Product	Quantity Ordered	Price Each	Order Date	City
0	12732	Keyboard	4	700	2023-01-01 00:00:00	Pune
1	19845	Laptop	2	700	2023-01-01 01:00:00	Mumbai
2	13264	Monitor	3	500	2023-01-01 02:00:00	Bangalore
3	14859	Phone	1	200	2023-01-01 03:00:00	Bangalore
4	19225	Monitor	4	100	2023-01-01 04:00:00	Mumbai

Next steps: Generate code with df New interactive sheet

```
# Convert date
df['Order Date'] = pd.to_datetime(df['Order Date'])

# Add Month column
df['Month'] = df['Order Date'].dt.month

# Add Sales column
df['Sales'] = df['Quantity Ordered'] * df['Price Each']

df.head()
```

	Order ID	Product	Quantity Ordered	Price Each	Order Date	City	Month	Sales
0	12732	Keyboard	4	700	2023-01-01 00:00:00	Pune	1	2800
1	19845	Laptop	2	700	2023-01-01 01:00:00	Mumbai	1	1400
2	13264	Monitor	3	500	2023-01-01 02:00:00	Bangalore	1	1500
3	14859	Phone	1	200	2023-01-01 03:00:00	Bangalore	1	200
4	19225	Monitor	4	100	2023-01-01 04:00:00	Mumbai	1	400

Next steps: Generate code with df New interactive sheet

```
#Grain 1: Best Month for Sales
monthly_sales = df.groupby('Month')['Sales'].sum()

best_month = monthly_sales.idxmax()
best_value = monthly_sales.max()

print("Best Month:", best_month)
print("Revenue:", best_value)
```

```
Best Month: 1
Revenue: 2220000
```

```
# Grain 2: Product Sold Most
product_sales = df.groupby('Product')['Quantity Ordered'].sum()

top_product = product_sales.idxmax()
print("Top Product:", top_product)
```

```
Top Product: Keyboard
```

```
#Grain 3: City with Highest Sales
city_sales = df.groupby('City')['Sales'].sum()

print(city_sales.sort_values(ascending=False))
```

```
City
Delhi        3865700
Bangalore    3794000
Pune         3683300
Mumbai       3646200
Name: Sales, dtype: int64
```

```
# Grain 4: Products Sold Together
# Simulate multiple products per order
df_dup = df[df.duplicated('Order ID', keep=False)]

grouped = df_dup.groupby('Order ID')['Product'].apply(lambda x: ','.join(x))

from collections import Counter
count = Counter()

for row in grouped:
    items = row.split(',')
    count.update(Counter([tuple(sorted(items))]))

print(count.most_common(5))
```

```
[(('Headphones', 'Monitor'), 173), (('Headphones', 'Keyboard'), 160), (('Monitor', 'Phone'), 152), (('Headphones', 'Phone'),
```

```
# Grain 5: Average Order Value
avg_order_value = df.groupby('Order ID')['Sales'].sum().mean()
print("Average Order Value:", avg_order_value)
```

```
Average Order Value: 2366.4666877170826
```

```
# Grain 6: Monthly Order Count
orders_per_month = df.groupby('Month')['Order ID'].count()
print(orders_per_month)
```

```
Month
1     1488
2     1168
3      744
4      720
5      744
6      720
7      744
8      744
9      720
10     744
11     720
12     744
Name: Order ID, dtype: int64
```

```
# Grain 7: Top Revenue Product
revenue_product = df.groupby('Product')['Sales'].sum()
print(revenue_product.sort_values(ascending=False))
```

```
Product
Keyboard      3064500
Headphones    3002700
Monitor       2995100
Laptop        2986900
Phone         2940000
Name: Sales, dtype: int64
```

```
#Grain 8: Average Price per Product
avg_price = df.groupby('Product')['Price Each'].mean()
print(avg_price)
```

```
Product
Headphones    591.926696
Keyboard      600.485437
Laptop        607.073791
Monitor       606.234414
Phone         596.002050
Name: Price Each, dtype: float64
```

```
#Grain 9: Quantity Sold per City
qty_city = df.groupby('City')['Quantity Ordered'].sum()
print(qty_city)
```

```
City
Bangalore    6402
Delhi        6378
Mumbai       6147
Pune         6034
Name: Quantity Ordered, dtype: int64
```

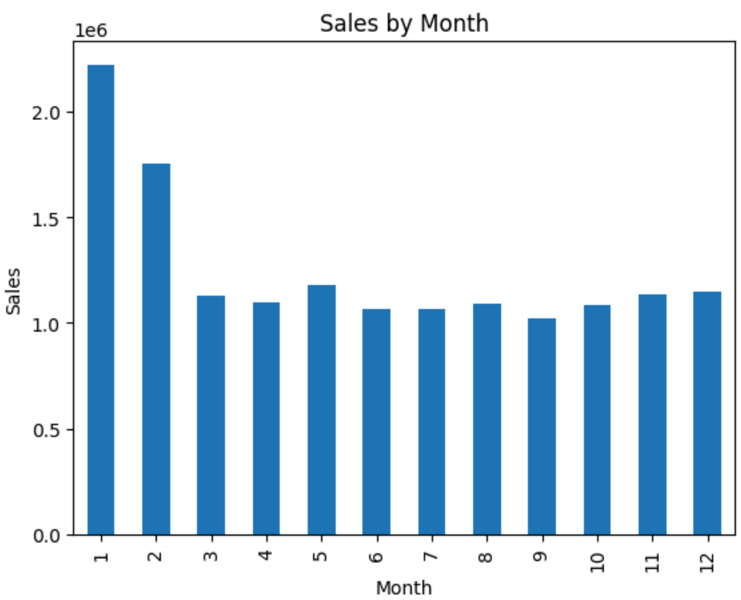
```
#Grain 10: Peak Sales Hour
df['Hour'] = df['Order Date'].dt.hour

hourly_sales = df.groupby('Hour')['Sales'].sum()
print(hourly_sales.idxmax())
```

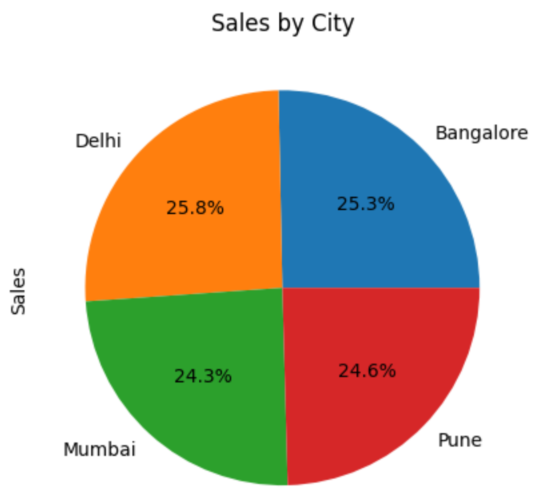
```
8
```

Visualizations

```
#Best Month Sales
monthly_sales.plot(kind='bar')
plt.title("Sales by Month")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```




```
#City Sales
city_sales.plot(kind='pie', autopct='%1.1f%%')
plt.title("Sales by City")
plt.show()
```




```
#Product Sales
product_sales.plot(kind='bar')
plt.title("Product Sales")
plt.show()
```